

Chapter 9

8 : If A , B , and C are sets, then $A - (B \cup C) = (A - B) \cup (A - C)$

LH		RH
$Proof : A - (B \cup C) \left\{ \begin{array}{l} \text{element } x \\ x \in A \\ x \notin B \\ x \notin C \end{array} \right.$	$(A - B) \cup (A - C)$	$\left\{ \begin{array}{l} \text{element } x \\ x \in A \quad \text{or} \quad x \in A \\ x \notin B \quad \text{or} \quad x \notin C \\ \text{makes} \\ x \in A \\ x \notin B \\ x \notin C \end{array} \right.$

Thus $LH = RH$ and the statement is true.

25 : For all $a, b, c \in \mathbb{Z}$, if $a \mid bc$, then $a \mid b$ or $a \mid c$.

Proof :

$$a = 4, b = 2, c = 6$$

$$bc = 12$$

$$4 \mid 12, 4 \nmid 2, 4 \nmid 6$$

Therefore, this statement is untrue.

Chapter 10

2 : Prove that $1^2 + 2^2 + 3^2 + 4^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$ for every positive integer n .

Proof :

$$\begin{array}{l|l}
 \text{Base Case : } n = 1 & n + 1 : n = 2 \\
 \hline
 1 = \frac{1(2)(3)}{6} & 1^2 + 2^2 = \frac{2(3)(5)}{6} \\
 1 = \frac{6}{6} & 5 = \frac{30}{6} \\
 1 = 1 & 5 = 5
 \end{array}$$

11 : Prove that $3 \mid (n^3 + 5n + 6)$ for every integer $n \geq 0$

Proof :

$$\begin{array}{l|l}
 \text{Base Case : } n = 0 & n + 1 : n = 1 \\
 \hline
 3 \mid (0 + 0 + 6) & 3 \mid (1 + 5 + 6) \\
 \text{True} & 3 \mid (12)
 \end{array}$$

Since S_n and S_{n+1} are true, then all $n \geq 0$ prove true for this statement.

16 : Prove $2^n + 1 \leq 3^n$ for all positive integers n .

Proof :

$$\begin{array}{l|l}
 \text{Base Case : } n = 1 & n + 1 : n = 2 \\
 \hline
 2 + 1 \leq 3 & 4 + 1 \leq 9 \\
 \text{True} & \text{True}
 \end{array}$$

Since S_n and S_{n+1} are true, then all positive numbers n prove true for this statement.

34 : Prove $3^1 + 3^2 + 3^3 + 3^4 + \dots + 3^n = \frac{3^{n+1}-3}{2}$ for every $n \in \mathbb{N}$

Proof :

<i>Base Case : $n = 1$</i>	<i>$n + 1 : n = 2$</i>
$3 = \frac{9-3}{2}$	$3 + 9 = \frac{27-3}{2}$
$3 = \frac{6}{2}$	$12 = \frac{24}{2}$
$3 = 3$	$12 = 12$
<i>True</i>	<i>True</i>

Since S_n and S_{n+1} are true, then all positive numbers n prove true for this statement.

$\forall a \in \mathbb{Z}, \exists b \in \mathbb{Z}, s.t. a = 4b + 3$

Proof :

$$a = 0$$

$$0 = 4b + 3$$

$$-3 = 4b$$

$$\frac{-3}{4} = b$$

$\frac{-3}{4}$ is not an integer, therefore this statement is untrue.

$\forall a \in \mathbb{Q}, \exists b \in \mathbb{Q}, s.t. a = 4b + 3$

Proof :

$$a = 4b + 3$$

$$a - 3 = 4b$$

$$\frac{a-3}{4} = b$$

Since a is a rational number, then $a - 3$ is a rational number. Therefore $\frac{a-3}{4}$ is a rational number.